

Nandish Jha

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EDUCATION

Bachelor of Science in Computer Engineering

Certificate in Professional Communication | Relevant: EE 221 (Analog Electronics), CME 342 (IC/SoC), CME 466 (Power/Tim

Bachelor of Science in Computer Science (In Progress)

Specialization in Computer Systems | Available to defer for full-time employment

University of Saskatchewan

Expected Apr. 2027

TECHNICAL SKILLS

Languages: Python, Verilog, SystemVerilog, C/C++, TCL, Bash, SQL

EDA & Hardware Tools: Cadence, Synopsys, QuestaSim, ModelSim, Quartus, Intel FPGA, Oscilloscopes, Logic Analyzers

Platforms: Linux, Git, Docker, Raspberry Pi SoC, RTOS (MicroC/OS-II)

Domains: IC Design Flow (RTL to Layout), SoC Architecture, Timing/Area/Power Analysis, Functional Verification, Technical Presentation & Communication

PROJECTS

Digital IC & SoC Design – CMOS Standard Cell Flow | *Cadence, Synopsys, Verilog, SystemC* 2022 – 2023

- Executed full IC design flow from RTL specification through physical layout using Cadence and Synopsys EDA toolchains; applied standard cell methodology across schematic, simulation, and layout stages
- Analyzed SoC bus structures and component interactions across multiple subsystem configurations; characterized and documented design behavior at each stage consistent with design review standards
- Developed platform-level understanding of component integration, delivery sequencing, and system architecture tradeoffs across simulation and physical implementation

Design of an Advanced Digital System | *Verilog, SystemVerilog, Cadence, Synopsys* 2024

- Applied Cadence and Synopsys toolchains to evaluate timing, area, and power tradeoffs across implementation alternatives; profiled component-level impact to inform system-level optimization decisions
- Developed automated scripting workflows for synthesis iteration, result extraction, and tradeoff comparison; structured findings into formal design review reports

UVM-Based Functional Verification – XSwitch | *SystemVerilog, UVM, Python, TCL, QuestaSim* 2023

- Developed complete UVM testbench achieving 95% functional coverage; applied constrained random stimulus and coverage-driven methodology to maximize fault detection across 200+ test scenarios
- Automated simulation compilation, execution, and coverage reporting using Python and TCL, reducing manual execution time by 80% and enabling fully reproducible regression runs
- Diagnosed design issues using assertion-based debugging; presented structured root cause analysis in academic design review sessions

IoT Smart System – SoC Platform | *Python, Raspberry Pi 4, TensorFlow, Linux* Jan. – Apr. 2024

- Built and deployed ML-driven IoT application on Linux-based SoC for real-time sensor data processing and automated decision-making; improved system efficiency by 25% through predictive analytics
- Implemented robust error handling and logging achieving 99% uptime in deployment; validated system behavior with stakeholders and iterated on improvements based on direct feedback

EXPERIENCE

Student Senator

Elected undergraduate representative; cross-functional policy collaboration

University of Saskatchewan Senate

Sep. 2026 – Apr. 2027

Campus Leadership

Led 2024 campaign (10% turnout increase); coordinated cross-functional team of volunteers

USSU Elections Campaign Manager

Front End / Cash Office

Cash operations, reconciliation, and customer resolution in high-volume environment

Real Canadian Superstore

2020 – 2021; 2024 – Present

CERTIFICATIONS

SoC Validation & Hardware Bring-up | UVM SystemVerilog Verification | AWS Certified Cloud Practitioner | AI/Deep Learning (Udemy)